

ULIP Integration Requirement Document

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Project Name: ULIP

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Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	12/09/2023	Girish Chander	Nayab Ali	Baseline Version
2.0	30/09/2024	Ayush Arya	Nayab Ali	Baseline Version

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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for authenticating Sarathi Response data with respect to driving license number and dob which we have received from sarathi system. This document details the following points:

1. **Data Integration and verification between SARATHI and ULIP web service:**
Identification of operational points where data integration and verification between SARATHI and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

User will share the information such as **driving license number and date of birth and will at least share either driver name or permit** based on data send by user ULIP system will call respective **SARATHI** API for accessing information and then verify data based on user input.

Following are the point for accessing vehicle information: -

1. This API will share driving license data based on information provided by user.
2. ULIP will expose API for getting data from SARATHI API and then verify the response if correct information is there or not in the response which we are getting from SARATHI API.
3. **SARATHI/verifySarathiData** API will take driving license and date of birth and will share at least driver name and permit from user and get data from SARATHI API and then will verify if correct driver name (which is **bioFullName** field in SARATHI API response) and permit (which is **covabbrv** field in SARATHI API response) is coming from SARATHI API response or not.

1.2 Data Integration between SARATHI System and ULIP application

User will share information to ULIP system based on data shared ULIP will verify the data for the user.

1.3 SARATHI/verifySarathiData

This API will take driving license number and connect with SARATHI API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	dlnumber	Driving license Number as available in SARATHI system.	(([A-Z]{2})(-)[0-9]{2}) ([A-Z]{2}[0-9]{2}) ((19 20)[0-9][0-9])[0-9]{7}\$	Required	16
2	dob	Date of birth as available in SARATHI system.	yyyy-mm-dd	Required	10
3	driverName	Driver name as available in SARATHI system.		Non Mandatory	
4	permit	Permit as available in SARATHI system.		Non Mandatory	

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **driving license number and date of birth and will at least share either driver name or permit** for which detail and verification needs to be done.
3. SARATHI API will share details of driving license information which exist in SARATHI system in JSON format.
4. ULIP system will verify the response which is received from SARATHI API based on the driver name and permit which is provided by the user.
5. ULIP system will provide the data in JSON Object format in response body.

Example:

Driving license Number **MP44 20100005566** , date of birth **1987-06-02**, driver name Vikas Tailor and **Permit** LMV is shared by ULIP system, then SARATHI-integration will make a request to SARATHI API.

Request:

The request must be of the following format, where driving license number and date of birth is passed within the URL body in JSON format.

For staging --

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03>

```
{
  "dlnumber": "MP44 20100005566",
```

```
{
  "dob": "1987-06-02",
  "driverName": "VIKAS TAILOR",
  "permit": "LMV", "MCWG"
}
```

Curl URL

```
curl --location --
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGluIiwiaWF0IjoxNjk1NzA2NjE3LCJhcHBzIjoiZGF0YXB1c2gifQ.inBggjzp9rdezG4JhkJtK0DnjdBt9G4ts6ubmk0YugIrnjnzYSnfP2E0qR0FHvGNxc2KS6o0GB3cwhWzEE0McQ' \
--header 'Cookie: __zlcmid=1DSlSDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-3b9c43a5b7c7' \
--data-raw '{
  "dlnumber": "MP44 20100005566",
  "dob": "1987-06-02",
  "driverName": "VIKAS TAILOR",
  "permit" : "LMV,MCWG"
}'
```

For production –

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03>

```
{
  "dlnumber": "MP44 20100005566",
  "dob": "1987-06-02",
  "driverName": "VIKAS TAILOR",
  "permit": "LMV", "MCWG"
}
```

Curl URL

```
curl --location --
request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGluIiwiaWF0IjoxNjk1NzA2NjE3LCJhcHBzIjoiZGF0YXB1c2gifQ.inBggjzp9rdezG4JhkJtK0DnjdBt9G4ts6ubmk0YugIrnjnzYSnfP2E0qR0FHvGNxc2KS6o0GB3cwhWzEE0McQ' \
```

```
--header 'Cookie: __zlcmid=1DSlSDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-3b9c43a5b7c7' \
--data-raw '{
  "dlnumber": "MP44 20100005566",
  "dob": "1987-06-02",
  "driverName": "VIKAS TAILOR",
  "permit" : "LMV,MCWG"
}'
```

For obtaining authentication token, refer [Section 1.4](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of DL does not exist in SARATHI system

```
{
  "response": [
    {
      "response": {
        "errorDesc": "No Response From Sarathi API"
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

b) In case Driver name verified successfully

Note: Driver name should be verified by **bioFullName** field in SARATHI API response.

```
{
  "responseStatus": "SUCCESS",
  "response": {
    "driverName": "Verified"
  }
}
```

c) In case Driver name not verified successfully

Note: Driver name should be verified by **bioFullName** field in SARATHI API response.

```
{
  "responseStatus": "SUCCESS",
  "response": {
    "driverName": "Verification failed"
  }
}
```

d) In case permit verified successfully

Note: PERMIT should be verified by **covabbrv** field in SARATHI API response.

```
{
  "responseStatus": "SUCCESS",
  "response": {
    "permitLmv": "Verified",
    "permitMCWG": "Verified"
  }
}
```

e) In case permit not verified successfully

Note: PERMIT should be verified by **covabbrv** field in SARATHI API response.

```
{
  "responseStatus": "SUCCESS",
  "response": {
    "permitLmv": "Verification Failed",
    "permitMCWG": " Verification Failed"
  }
}
```

Status Code 200: vehicle data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.4 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in SARATHI system through a data exchange URL that has following form:

For staging –

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03>

Body:

```
{
  "dlnumber": "GJ04 20120005008",
  "dob": "1987-05-26",
  "driverName": "VIKAS TAILOR",
}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Brief summary of HTTP Basic Authentication is as follows:

First time when User will hit ULIP secured API through username and password for access token as follow-

```
curl --location --request POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/Login' \
  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "username": "xx",
    "password": "xx"
  }'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImIhdCI6MTYzMDU2MTc1NywiYXBwcyI6ImRh
dGFwdXNoIn0.06h4916NwL81GPvjFDUgeEHLnMWhAh_NdW3okMOa6Q2ddw89RWmdDvDx4UYRoUeJ8
yfPZYNBxhTPoQZ-uUzgTg'
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which

represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for vehicle details as follow -

```
curl --location --
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0Ij
oxNjk1NzA2NjE3LCJhcHBzIjoizGF0YXB1c2gifQ.inBggjSp9rdezG4JhkJtK0DnjdBt9G4ts6ub
mk0YugIrnjnzYSnfP2E0qR0FHvGNxc2KS6o0GB3cwhWzEEOMcQ' \
--header 'Cookie: __zlcmid=1DSLSDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-
d0d1-3b9c43a5b7c7' \
--data-raw '{
  "dlnumber": "MP44 20100005566",
  "dob": "1987-06-02",
  "driverName": "VIKAS TAILOR",
  "permit": "LMV,MCWG"
}'
```

For production –

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03>

Body:

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  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "username": "xx",
    "password": "xx"
  }'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImhhdCI6MTYzMDU2MTc1NywiYXBwcyI6ImRh
dGFwdXNoIn0.06h4916NwL81GPvjFDUgeEHLnMWhAh_NdW3okMOa6Q2ddw89RWmdDvDx4UYRoUeJ8
yfPZYNBxhTPoQZ-uUzgTg'
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

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```
curl --location --
request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/03' \
  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGJwIiwiaWF0Ij
oxNjk1NzA2NjE3LCJhcHBzIjoizGF0YXB1c2gifQ.inBggj9p9rdezG4JhkJtK0DnjdBt9G4ts6ub
mk0YugIrnjnzYSnfP2E0qR0FHvGNxc2KS6o0GB3cwhWzEE0McQ' \
  --header 'Cookie: __zlcid=1DSLSDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-
d0d1-3b9c43a5b7c7' \
  --data-raw '{
    "dlnumber": "MP44 20100005566",
    "dob": "1987-06-02",
    "driverName": "VIKAS TAILOR",
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